

Design of a Lightweight Crop Disease Detection and Decision Support System for South Asia with Out-of-Distribution Detection and Verified Treatment Recommendations*

Project Report

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*Source code is available at: https://github.com/myselfabk5/crop_disease_detection

Abstract

Timely and correct diagnosis of crop disease is a persistent bottleneck for smallholder farmers across South Asia, where access to trained agricultural extension officers is limited and unverified advice circulating online is common. This report describes a course project that builds an image-based crop disease diagnosis pipeline aimed at addressing three concerns simultaneously: recognition accuracy, awareness of the model’s own competence boundary, and reliability of the treatment advice that follows a prediction. The system combines an EfficientNet-B0 classifier fine-tuned on a 94-class South Asian crop disease dataset, a convolutional autoencoder that flags inputs likely to fall outside the training distribution using reconstruction error, Grad-CAM based visual explanations, and a static, source-attributed treatment knowledge base rather than free-text generation. The classifier attains 89.45% test accuracy, and the autoencoder-based out-of-distribution (OOD) detector achieves an AUROC of 0.8183 in separating in-distribution validation reconstruction errors from out-of-distribution reconstruction errors. The complete pipeline is deployed as an interactive Streamlit [\[7\]](#) application. The report documents the design rationale, architecture, quantitative results, and limitations of the system.

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1. Introduction

Plant disease is one of the principal causes of yield loss in South Asian agriculture, and the loss is compounded not by the biology of the disease itself but by the time it takes a farmer to recognise it. A large share of the region’s cultivators work on small, geographically dispersed plots and cannot economically justify a visit from a trained plant pathologist for every suspicious leaf. In the absence of expert access, farmers frequently turn to informal, crowd-sourced sources of agricultural advice, which are not curated for accuracy and can lead to inappropriate or even harmful treatment choices. The consequence is a pattern that repeats across crops and seasons: disease is noticed late, spreads before it is controlled, and reduces both yield and income.

Deep learning based image classification [1] offers a plausible partial remedy, since a smartphone photograph of an affected leaf can, in principle, be classified as quickly as it can be taken. However, a classifier trained on a fixed set of crop-disease categories is only reliable within that set [2]; if the input image depicts something the model was never trained to recognise — a different crop species, a non-plant object, or a badly degraded photograph — a standard softmax classifier will still output one of its known classes. This is a serious problem for a decision-support tool that is meant to be trusted by a non-expert user. A second, related problem is that even a correct diagnosis is not by itself actionable; the farmer additionally needs treatment guidance, and if that guidance is produced by an open-ended generative model, it is vulnerable to hallucination in exactly the situations — rare diseases, ambiguous symptoms — where an incorrect answer is least likely to be caught by the user.

This project therefore treats crop disease diagnosis as a three-part problem rather than a single classification task: (i) classify the disease when the input is recognisable, (ii) explicitly detect and flag inputs that are not recognisable, and (iii) attach treatment information that is retrieved from a fixed, source-attributed record rather than generated on the fly.

The remainder of the report is organised as follows. Section 2 states the problem more formally. Section 3 describes the dataset used for classification. Section 4 presents the methodology, covering the CNN classifier, the transfer-learning strategy, the EfficientNet-B0 backbone, the autoencoder, the OOD detection rule, and the treatment recommendation module. Section 5 describes the experimental and deployment setup. Section 6 reports the quantitative results. Sections 7 & 8 discuss the results, the limitations of the current system, directions for future work, and concludes the report.

2. Problem Statement

Given a photograph of a crop leaf supplied by a non-expert user, the system is required to satisfy five design requirements simultaneously.

First, the system must classify the crop-disease pair present in the image with high top-1 accuracy across a fine-grained label space of 94 classes spanning 14 crop species. Second, the system must estimate whether the input image is likely to belong to a distribution the classifier was not trained on, so that a prediction is not silently returned for content the model cannot meaningfully interpret. Third, when an input is judged to be out-of-distribution, the system must avoid presenting the resulting prediction as though it were reliable, and should instead surface this uncertainty to the user. Fourth, whenever a prediction is returned, the system must attach a treatment recommendation whose content can be traced to a named external source, rather than one synthesised on demand by a language model. Fifth, the overall system, and in particular its classification backbone, must remain lightweight enough that it is plausible to deploy the model using small compute rather than requiring server-grade compute at inference time.

These requirements are in tension with one another to varying degrees — for instance, a stricter OOD threshold reduces the risk of acting on unreliable predictions but increases the rate at which genuine in-distribution images are unnecessarily flagged — and Section 6 quantifies where the current implementation sits on this trade-off.

3. Dataset

The classifier was trained on the Bangladesh Multi-Crop Disease Classification Dataset, a publicly available image corpus hosted on Hugging Face [5]. The dataset was developed to support automated plant disease identification for crops commonly cultivated in Bangladesh and the wider South Asian region. It combines several publicly available agricultural image datasets into a unified benchmark, covering diseases that are economically important to regional farming systems.

The complete dataset contains 123,588 images belonging to 94 classes across 14 crop species. For computational efficiency, a subset of the Dataset is used for the transfer learning. The selected subset contains 31,002 images while preserving the original class structure of 94 classes across 14 crop species. Images are divided into predefined training, validation, and testing subsets, as summarized in Table 1.

The dataset covers the following crops: banana, cauliflower, corn, cotton, guava, jute, mango, papaya, potato, rice, sugarcane, tea, tomato, and wheat. Together these represent many of the

Table 1: Dataset split used for model development

Split	Images	Percentage
Training	23,104	74%
Validation	4,104	13%
Testing	3,794	13%
Total	31,002	100%

major food and cash crops cultivated throughout South Asia.

Every crop includes at least one healthy class. The inclusion of healthy samples is essential for reducing false positive disease predictions and provides an appropriate outcome for the treatment recommendation module described in Section 4.6.

The dataset was created by integrating multiple publicly available agricultural image repositories, including datasets hosted on Hugging Face and the Mendeley Data Repository, thereby providing a diverse collection of disease images captured under varying acquisition conditions. This diversity improves the robustness of the trained classifier and better reflects the variability encountered in practical agricultural environments.

4. Methodology

The deployed system consists of two neural networks trained independently of one another — a classifier and an autoencoder — plus a deterministic post-processing stage that has no learned parameters. This section describes each component as implemented, together with the reasoning behind the corresponding design choice.

4.1 CNN Classifier

The classifier receives an image, resized to 224×224 pixels and normalised with ImageNet channel statistics (mean $[0.485, 0.456, 0.406]$, standard deviation $[0.229, 0.224, 0.225]$), matching the normalisation of the EfficientNet-B0. The network outputs a 94-way logit vector, which is converted to class probabilities with a softmax and to a single prediction by taking the arg max; the corresponding maximum probability is reported to the user as the model’s confidence.

4.2 Transfer Learning

The classifier is built on top of an ImageNet-pretrained EfficientNet-B0 [4] [6] rather than a network trained from randomly initialised weights. The motivation is straightforward: a 94-class, moderately sized agricultural image corpus is small relative to the scale of data typically

required to learn general-purpose visual features — edges, textures, colour gradients, simple shapes from scratch, whereas ImageNet pretraining already encodes exactly these general features. Reusing them lets the fine-tuning stage concentrate its limited labelled data on the part of the problem that is actually novel, namely mapping leaf-level texture and colour patterns onto disease categories, rather than relearning low-level vision from zero. In practice this is implemented by replacing the original 1000-way ImageNet classification head with a new head sized for 94 outputs and continuing training on the crop disease corpus.

4.3 EfficientNet-B0

EfficientNet-B0 [6] was selected as the backbone specifically because of the project’s lightweight deployment target. Tan and Le [6] observed that convolutional networks are conventionally scaled up along a single dimension — more layers, wider layers, or higher input resolution — in isolation, and showed that jointly scaling all three dimensions by a fixed compound coefficient yields a family of networks that are simultaneously more accurate and more parameter-efficient than networks scaled along one dimension alone [6]. EfficientNet-B0, the base member of this family found via neural architecture search, is built from mobile inverted bottleneck convolution (MBConv) blocks with squeeze-and-excitation attention, and is small enough — on the order of 5 million parameters — to be a realistic candidate for a lightweight model, in contrast to substantially larger architectures such as ResNet-50 or VGG-16. Concretely, the implementation exposes the backbone’s stages as a sequential list of blocks, and the explainability module uses the final element of this list as the target convolutional layer for Grad-CAM [3], since it is the last point in the network at which spatial location information is still present before global pooling and classification.

4.4 Autoencoder

The Out of Distribution (OOD) detector is a plain convolutional autoencoder, trained separately from the classifier and used purely to produce a scalar reconstruction-error score.

The encoder consists of four convolutional layers with 32, 64, 128, and 256 output channels respectively, each using a 3×3 kernel, stride 2, and padding 1, with a ReLU activation after every layer. Each stride-2 convolution halves the spatial resolution, so a 224×224 input is reduced to 14×14 after four such layers, since 224 divided by 2 four times is 14. The resulting $256 \times 14 \times 14$ feature map is flattened and projected by a fully connected layer to a 32-dimensional latent vector.

$$z = f_{\text{enc}}(x), \quad z \in \mathbb{R}^{32}, \quad x \in \mathbb{R}^{3 \times 224 \times 224} \quad (1)$$

The decoder mirrors the encoder: a fully connected layer expands the latent vector back to $256 \times 14 \times 14$, which is reshaped and passed through four transposed convolutions (kernel 4, stride 2, padding 1) that progressively reduce the channel count from 256 to 128, 64, 32, and finally 3, each followed by a ReLU except the last, which uses a sigmoid activation to bound every output pixel to the $[0, 1]$ range.

$$\hat{x} = f_{\text{dec}}(z), \quad \hat{x} \in [0, 1]^{3 \times 224 \times 224} \quad (2)$$

The sigmoid output range is why the autoencoder’s input transform resizes the image and converts it to a tensor without any ImageNet normalisation, unlike the classifier’s input transform: the reconstruction target is the raw $[0, 1]$ pixel intensity, and normalising the input while leaving the sigmoid-bounded output un-normalised would make the two incomparable. This is a small but deliberate implementation detail that keeps the reconstruction loss meaningful.

Reconstruction quality is measured with per-pixel mean squared error, averaged over the channel, height, and width dimensions of a single image to obtain one scalar loss value:

$$L(x) = \frac{1}{C \cdot H \cdot W} \sum_{c,h,w} (x_{c,h,w} - \hat{x}_{c,h,w})^2 \quad (3)$$

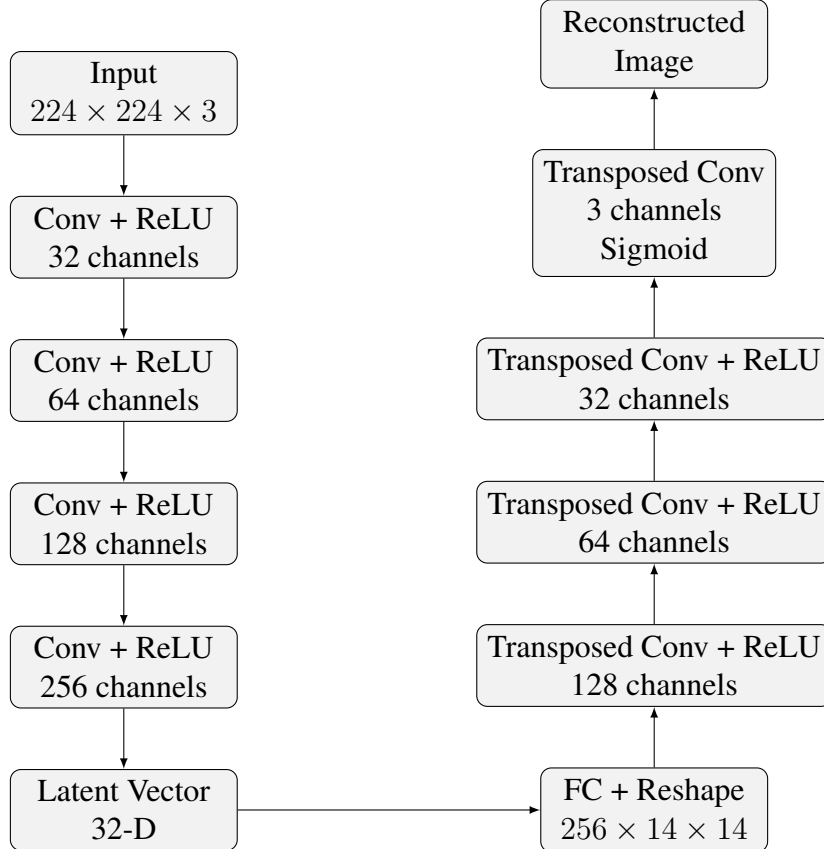


Figure 1: Architecture of the convolutional autoencoder used for out-of-distribution detection

4.5 OOD Detection

The intuition behind using reconstruction error for OOD detection is that an autoencoder trained only on in-distribution images can only compress and reconstruct the visual patterns it has seen; presented with an image drawn from a different distribution — an unrelated object, a different kind of scene, or an unrecognisable or heavily degraded photograph — the decoder has no learned basis for reconstructing it faithfully, and the reconstruction error rises accordingly. The deployed decision rule compares the scalar loss $L(x)$ against a fixed threshold τ , currently set to 0.01:

$$\text{decision}(x) = \begin{cases} \text{“not proceed”} & \text{if } L(x) > \tau \\ \text{“proceed”} & \text{otherwise} \end{cases} \quad \tau = 0.01 \quad (4)$$

This operating point sits between the reported validation-set mean reconstruction error of 0.0099 and the reported OOD-set mean of 0.0209 — specifically, which is consistent with a threshold chosen to reject a majority of OOD distribution images. When an image is flagged, the Streamlit interface described in Section 5 does not withhold the prediction outright; instead it surfaces the reconstruction error to the user and requires an explicit confirmation before the prediction is shown, which keeps a human in the loop for the cases the automated check is least confident about.

4.6 Treatment Recommendation Module

Once a class index is produced by the classifier, the treatment module performs a two-step deterministic lookup rather than any form of text generation. The predicted class index is first mapped to a human-readable class key using the label mapping file, and that key is then used to retrieve a fixed record from a static knowledge base containing, for every one of the 94 classes, the crop name, a symptom description, a probable cause, an organic treatment, a chemical treatment, a prevention strategy, a qualitative severity rating, and an information source with its URL [8] [9]. Because the retrieved text is copied verbatim from this pre-authored record and never produced by a generative model conditioned on the image or the prediction, the module cannot fabricate a treatment that does not already exist in the knowledge base; the only failure mode available to it is retrieving the wrong — but still internally consistent and sourced — record, which happens if and only if the upstream classification is itself not correct.

Finally, the retrieved record and the classifier’s confidence score are passed to a presentation layer that renders one of ten differently formatted (but content-identical) textual report templates, chosen uniformly at random at render time. This randomisation affects only cosmetic formatting — headings, ordering of fields, use of bullet markers — and has no bearing on the

underlying agronomic content, which is fixed by the knowledge base lookup described above.

5. Experimental Setup

The system is deployed as an interactive Streamlit application with three explicit states: awaiting an image upload, an OOD warning-and-confirmation state, and a results state. A user supplies an image either from a file or a device camera; the image is immediately passed through the autoencoder-based OOD check described in Section 4.5. If the reconstruction loss is below or equal to the threshold, the application proceeds directly to the classification pipeline. If it is above the threshold, the application interrupts the flow, displays the measured reconstruction error together with an explicit warning that the forthcoming prediction may be unreliable, and requires the user to choose to proceed before any prediction is computed and shown; choosing not to proceed simply resets the session for a new upload.

When the pipeline does run, it executes, in order, the classifier described in Section 4.1, Grad-CAM explainability computed against the final block of the EfficientNet-B0 backbone, and the knowledge-base lookup described in Section 4.6, before rendering one of the ten textual report templates and offering it as a downloadable text file.

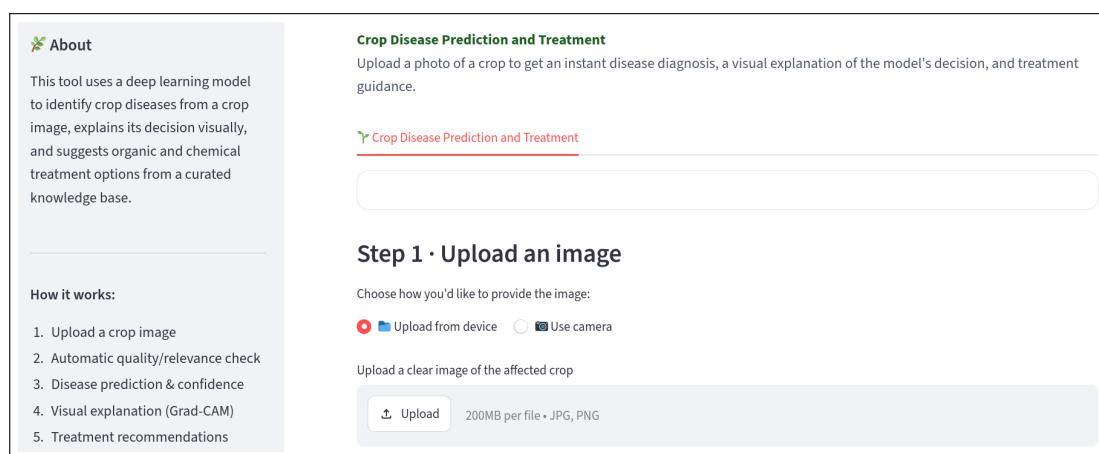


Figure 2: Streamlit dashboard

6. Results

6.1 Classification Performance

The fine-tuned EfficientNet-B0 classifier achieves a test accuracy of 89.45% across the 94-class label space. Given that the label space includes crops represented by nine or ten fine-grained disease categories. An accuracy in the high-eighties is a reasonable indication that

the transferred ImageNet features generalise usefully to leaf- and fruit-level symptom textures, without implying the task is easy. The remaining error, on the order of one in ten test images, is most plausibly concentrated among disease pairs that share a similar visual signature. Image quality effects such as inconsistent lighting, background clutter, and camera focus, all of which are characteristic of field photography as opposed to the controlled conditions typical of earlier plant-disease corpora [1], are also a plausible contributor to the residual error.

Table 2: Performance of the EfficientNet-B0 classifier

Metric	Value (%)
Validation Accuracy	89.64
Test Accuracy	89.45

6.2 Out-of-Distribution Detection

The autoencoder-based OOD detector is characterised by the following four reconstruction-loss statistics and one ranking metric, reported below:

Table 3: Reconstruction-loss statistics and OOD ranking metric

Quantity	Value
Validation mean reconstruction loss	0.0099
Validation standard deviation	0.0085
OOD mean reconstruction loss	0.0209
OOD standard deviation	0.0095
AUROC	0.8183

The reported statistics are based on 100 in-distribution samples and 100 out-of-distribution (OOD) samples. The OOD mean reconstruction loss is a little over twice the validation mean, and the gap between the two means (0.0109) is roughly 1.28 validation standard deviations, which is consistent with the qualitative expectation from Section 4.5 that out-of-distribution content reconstructs less faithfully than in-distribution leaf imagery. However, the two standard deviations are of comparable magnitude to this gap, which means the two loss distributions overlap rather than being cleanly separated; a fixed scalar threshold therefore cannot achieve perfect discrimination, and this is exactly what the AUROC value reflects. An AUROC of 0.8183 means that, for a randomly drawn in-distribution and a randomly drawn out-of-distribution image, the OOD image receives the higher reconstruction error roughly 82% of the time — a substantial improvement over the 50% expected from chance, but well short of the 1.0 that would indicate perfect separability. In practical terms, at any fixed operating threshold some genuinely in-distribution images will be flagged unnecessarily and some genuinely

out-of-distribution images will be passed through as though they were reliable; the reported statistics do not claim, perfect OOD detection.

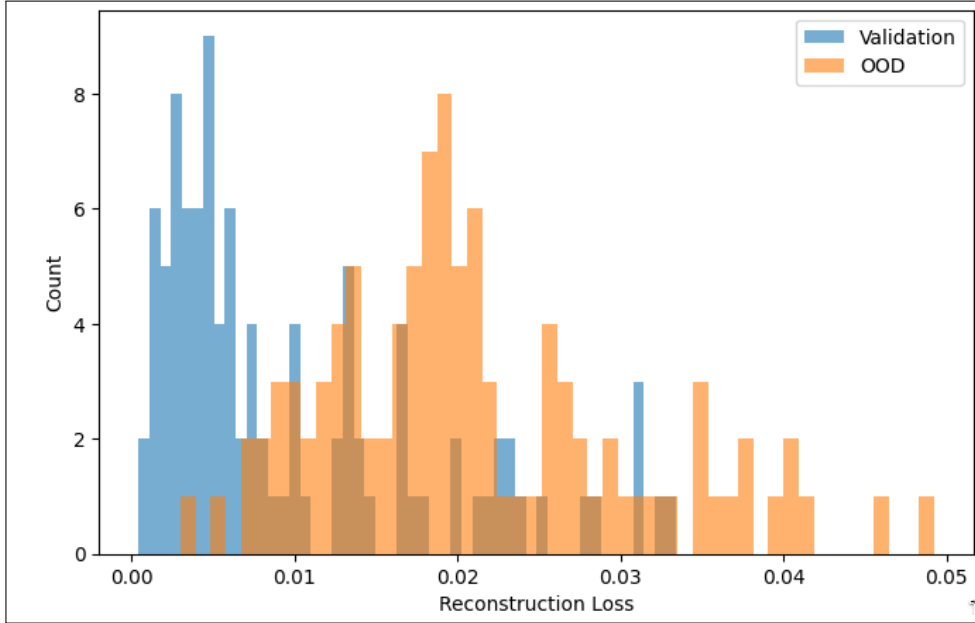


Figure 3: Reconstruction loss distribution

7. Discussion

The results demonstrate that the proposed system successfully meets the primary design objectives outlined in Section 2. The fine-tuned EfficientNet-B0 classifier achieves a test accuracy of 89.45% across 94 crop disease classes while maintaining a lightweight architecture suitable for deployment on resource-constrained systems. Considering the fine-grained nature of the classification task, where several diseases exhibit highly similar visual symptoms across multiple crop species, this level of performance indicates that transfer learning from ImageNet provides an effective feature representation for agricultural image classification.

The addition of the autoencoder-based out-of-distribution (OOD) detector extends the system beyond conventional crop disease classifiers. Rather than always producing a prediction, the system first evaluates whether an input image is sufficiently similar to the training distribution. The obtained AUROC of 0.8183 indicates that reconstruction error provides a useful signal for distinguishing familiar crop images from previously unseen inputs. Although the separation between in-distribution and OOD samples is not perfect, the detector substantially reduces the likelihood that obviously unfamiliar images are treated with the same confidence as known crop diseases. This additional safety mechanism improves the practical reliability of the overall decision-support pipeline.

An important strength of the proposed system is that classification, uncertainty estimation,

explainability, and treatment recommendation operate together within a single workflow. Grad-CAM visualisations allow users to verify that predictions are based on disease-affected regions of the leaf rather than background artefacts, improving the interpretability of model decisions. Similarly, the treatment recommendation module retrieves information from a static, source-attributed knowledge base instead of generating recommendations using a language model. This design choice eliminates the possibility of fabricated treatment advice and ensures that every recommendation can be traced back to a documented source.

The experimental results also demonstrate good generalisation. The validation accuracy (89.64%) and test accuracy (89.45%) are nearly identical, suggesting that the model does not suffer from significant overfitting despite being trained on a moderately sized subset of the original dataset. This indicates that the transfer learning strategy and EfficientNet-B0 architecture together provide an effective balance between model capacity and generalisation performance.

Nevertheless, the reported results also identify opportunities for further improvement. The overlap between the reconstruction-error distributions of in-distribution and OOD samples means that some unfamiliar inputs may still be accepted while some valid crop images may be unnecessarily flagged. Likewise, classification errors remain inevitable for visually similar diseases or images captured under challenging field conditions such as poor illumination, occlusion, or motion blur. These limitations are expected for practical agricultural datasets and motivate future work on more advanced OOD detection techniques and larger, more diverse training datasets.

Overall, the proposed system represents a practical improvement over conventional crop disease classifiers by combining accurate disease recognition with uncertainty estimation, visual explainability, and trustworthy treatment recommendations within a lightweight deployment framework. Rather than focusing solely on maximising classification accuracy, the project emphasises producing predictions whose reliability can be better assessed by end users, making the system more suitable for real-world agricultural decision support.

8. Conclusion

This project presented the design and implementation of a lightweight crop disease diagnosis system intended for practical agricultural decision support in South Asia. Unlike conventional image classification pipelines that focus solely on disease recognition, the proposed system integrates four complementary components: a deep learning classifier for disease identification, an autoencoder-based out-of-distribution (OOD) detector for uncertainty estimation, Grad-CAM visualisations for model interpretability, and a source-attributed treatment recommendation module. Together, these components provide not only a prediction but also supporting information that enables users to better assess the reliability of the system’s output.

The EfficientNet-B0 classifier achieved a validation accuracy of 89.64% and a test accuracy of 89.45% across 94 crop disease classes while maintaining a lightweight architecture suitable for deployment on resource-constrained devices. The convolutional autoencoder further improved the robustness of the system by distinguishing in-distribution and out-of-distribution inputs with an AUROC of 0.8183, allowing potentially unreliable predictions to be identified before being presented to the user. The integration of Grad-CAM enables visual inspection of the image regions influencing each prediction, while the deterministic treatment recommendation module provides transparent, source-attributed guidance without relying on generative language models.

Although the current implementation does not eliminate all classification or OOD detection errors, the results demonstrate that combining prediction accuracy with uncertainty estimation and explainability produces a more reliable decision-support system than a conventional classifier alone. The small gap between validation and test accuracy also indicates that the transfer learning strategy generalises well to unseen images despite the fine-grained nature of the classification task.

Overall, the proposed system demonstrates that reliable crop disease diagnosis requires more than high classification accuracy. By combining lightweight deep learning models with uncertainty awareness, visual explainability, and trustworthy treatment recommendations, this work provides a practical framework for AI-assisted agricultural support. With further expansion of the dataset, improved OOD detection techniques, and expert validation of the treatment knowledge base, the proposed approach has the potential to evolve into a dependable tool for supporting farmers and agricultural extension services in real-world environments.

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